Solving for angles in right triangles

1 Find θ for each of the following. Round your answer to the nearest degree.

a $\cos\theta = 0.146$

b $\tan\theta = 0.21$

c $\sin\theta = 0.825$

d $\tan\theta = 0.575$

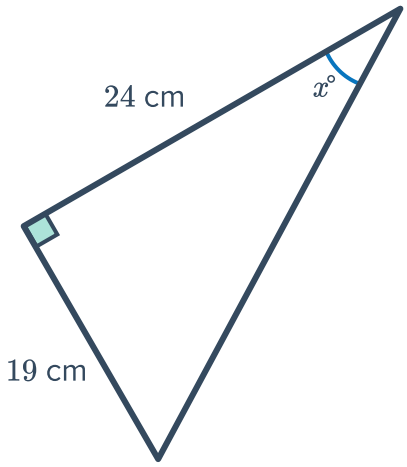
e $\cos\theta = 0.706$

f $\tan\theta = 1.2$

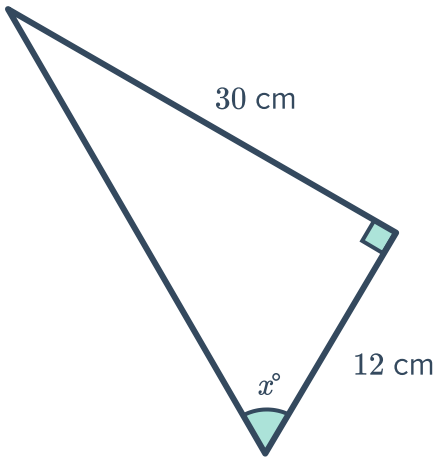
g $\tan\theta = \dfrac{9}{5}$

2 For each of the following diagrams, use the tangent ratio to find the measure of the angle marked with a variable, correct to the nearest degree.

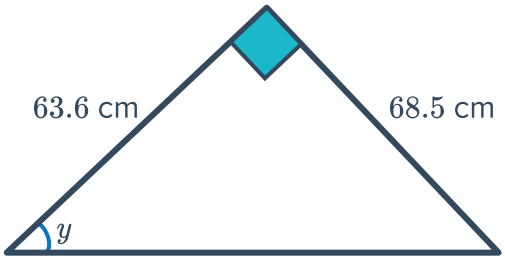
a



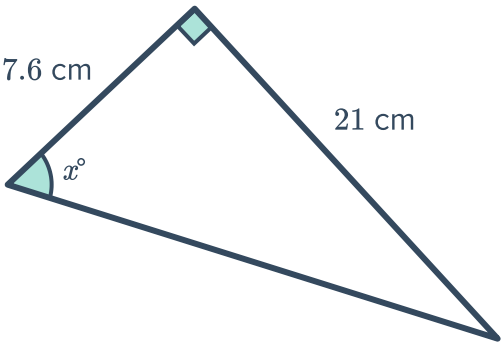
b



c

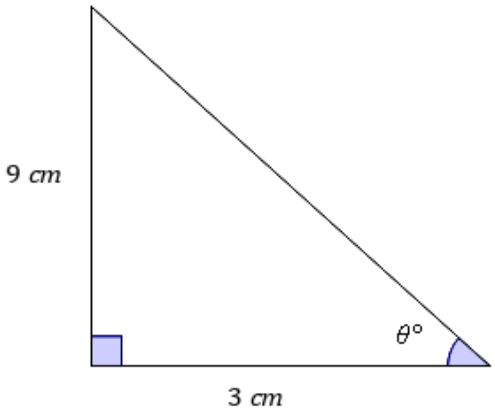


d

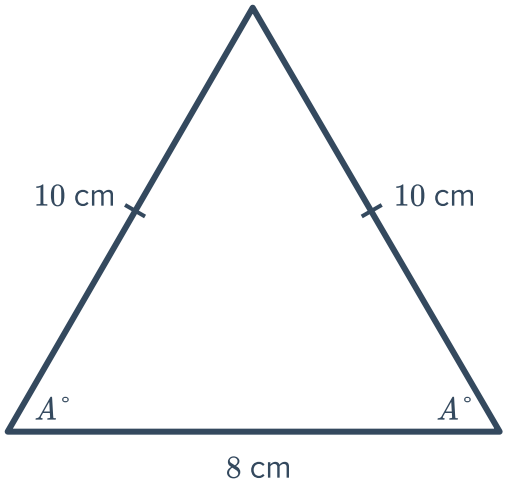


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3 By considering the given sides of the triangle, determine whether θ is less than or greater than 45°.



4 An isosceles triangle has the equal side lengths of 10 cm and a base of 8 cm. Calculate A to one decimal place.



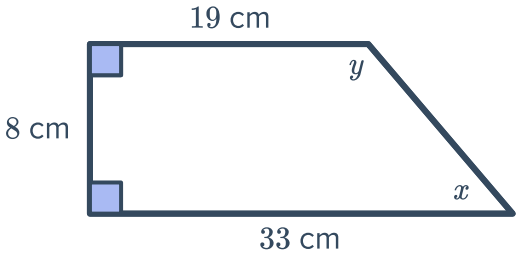
5 In $\triangle MNP$, $\angle MNP = 90^\circ$, $MN = 18$ and $NP = 17$. If the measure of $\angle MPN$ is t°, find t correct to the nearest degree.

6 The lengths of the opposite and adjacent sides of angle measuring x in a right triangle can be written in the ratio $4 : 3$. Solve for x correct to the nearest degree.

7 Consider the following diagram.

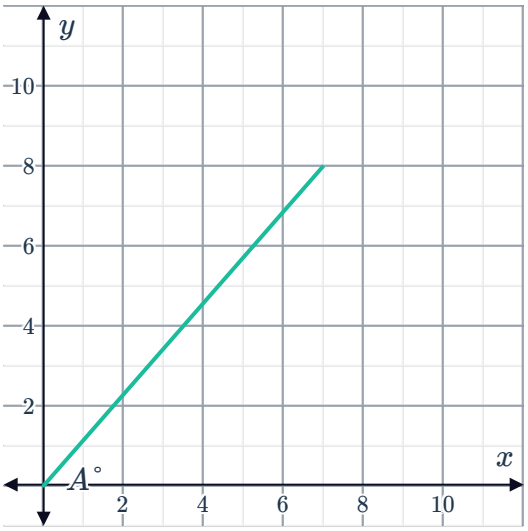
a Find x, correct to two decimal places.

b Find y, correct to two decimal places.



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8 The straight line segment makes an angle of A° with the positive x-axis. Find A, correct to the nearest degree.



9 Given that $\sin\theta = \frac{1}{2}$, find the value of the following:

a θ **b** $\cos\theta$ **c** $\tan\theta$

10 Given that $\tan\theta = \frac{1}{\sqrt{3}}$, find the value of the following:

a θ **b** $\sin\theta$ **c** $\cos\theta$

11 For each of the of the following trigonometric ratio, find:

i θ **ii** $\sin\theta$

a $\cos\theta = \frac{\sqrt{3}}{2}$ **b** $\tan\theta = 1$ **c** $\tan\theta = \frac{\sqrt{3}}{3}$

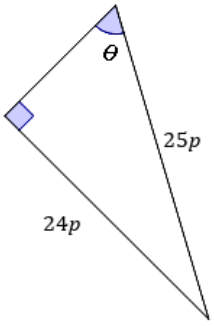
12 For each of the following trigonometric ratio, find:

i θ **ii** $\tan\theta$

a $\cos\theta = \frac{1}{2}$ **b** $\cos\theta = \frac{1}{\sqrt{2}}$ **c** $\sin\theta = \frac{\sqrt{3}}{2}$

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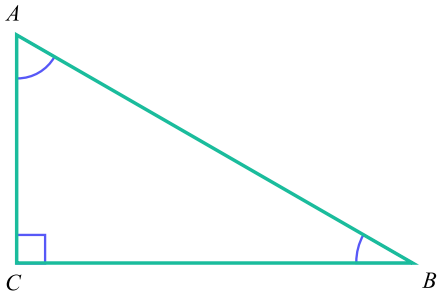
13 Find the value of $\tan\theta$ in this triangle.



14 Consider $\triangle ABC$.
Determine whether the following statements are true.

a $\cos A = \sin B$ **b** $\cos A = \sin A$

c $\sin A = \sin B$ **d** $\cos A = \cos B$

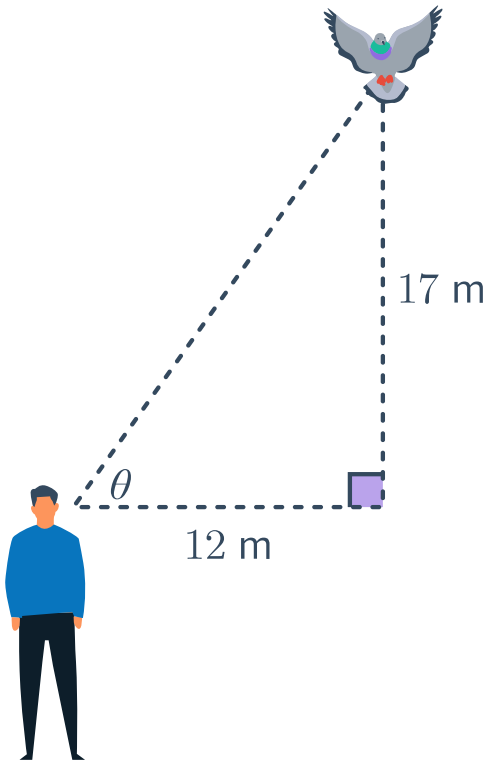


15 In the figure, a string of lights joins the top of the tree to a point on the ground 23.9 m away.
If the angle the string of lights would make with the ground is θ°, find θ to two decimal places.

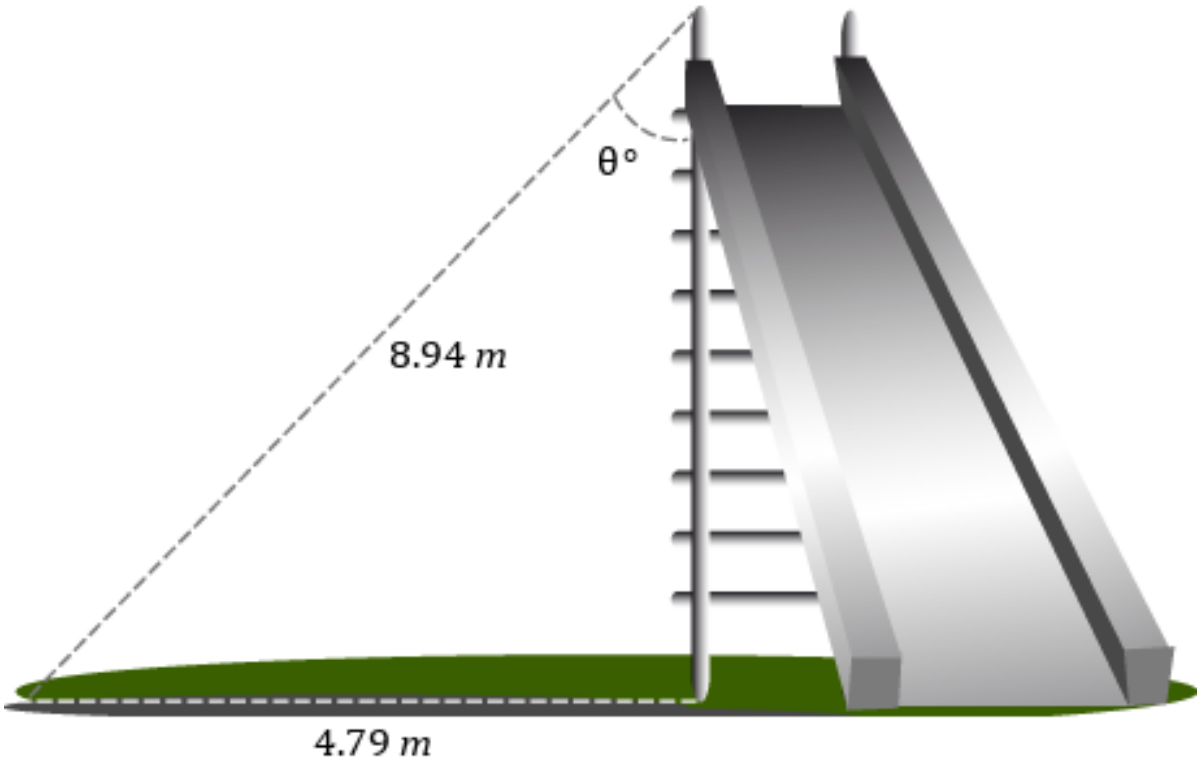


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16 The person in the figure sights a pigeon above him.
If the angle the person is looking at is θ, find θ in degrees. Round your answer to two decimal places.

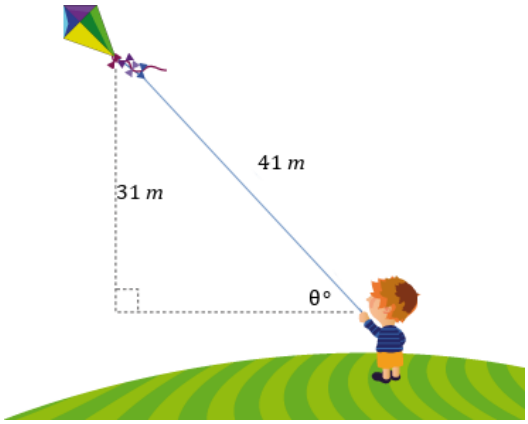


17 In the figure, the slide casts a shadow 4.79 m along the ground. The distance between the tip of the shadow and the top of the slide is 8.94 m.
Find the value of θ correct to two decimal places.

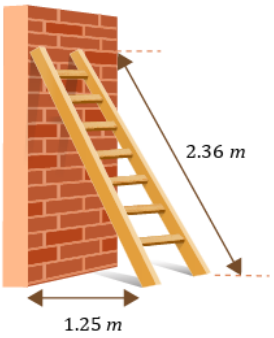


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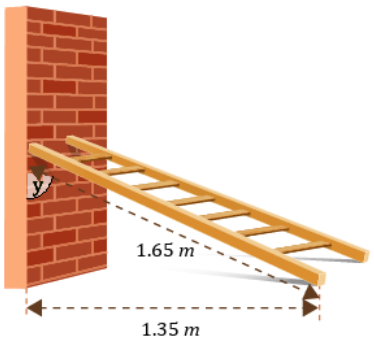
18 A boy flying his kite releases the entire length of his string which measures $41\ \text{m}$, so that the kite is $31\ \text{m}$ above him.
If the angle the string makes with the horizontal ground is θ°, find θ to two decimal places.



19 A ladder measuring $2.36\ \text{m}$ in length is leaning against a wall.
If the angle the ladder makes with the ground is y°, find y to two decimal places.

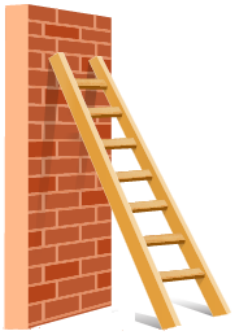


20 A ladder measuring $1.65\ \text{m}$ in length is leaning against a wall.
If the angle the ladder makes with the wall is y°, find y to two decimal places.



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21 A ladder of length $3.46\ \text{m}$ needs to make an angle with the wall of between 10° and 20° for it to be safe to stand on. At steeper angles, the ladder is at risk of toppling backwards when the climber leans away from it and at shallower angles, the ladder may lose its grip on the ground.



a If the base of the ladder is $190\ \text{cm}$ from the wall, and the angle the ladder makes with the wall is x°, find x to two decimal places.

b If the base of the ladder is $42\ \text{cm}$ from the wall, and the angle the ladder makes with the wall is y°, find y to two decimal places.

c If the base of the ladder is $86\ \text{cm}$ from the wall, and the angle the ladder makes with the wall is z°, find z to two decimal places.

d Hence, which of the three lengths from the wall is the ladder safe?

22 A ship dropped anchor off the coast of a resort. The anchor fell $73\ \text{m}$ to the sea bed. During the next 5 hours, the ship drifted $182\ \text{m}$.
Calculate x, the measure of the angle between the anchor line and the surface of the water. Round your answer to the nearest degree.

Additional questions

23 Find the value of the following expressions, rounding your answer to the nearest degree:

a $\sin^{-1}(0.7035)$ b $\cos^{-1}(0.4447)$ c $\tan^{-1}(0.5085)$ d $\sin^{-1}(0.2734)$

e $\cos^{-1}(0.6484)$ f $\tan^{-1}(8.1519)$ g $\sin^{-1}(0.8176)$ h $\cos^{-1}(0.7035)$

24 For each of the following, find θ to the nearest degree:

a $\tan\theta = 0$ b $\tan\theta = 1$ c $\cos\theta = 0.5$ d $\sin\theta = 0.906$

e $\tan\theta = 2.2$ f $\sin\theta = \dfrac{43}{47}$ g $\cos\theta = \dfrac{31}{53}$ h $\tan\theta = \dfrac{41}{31}$

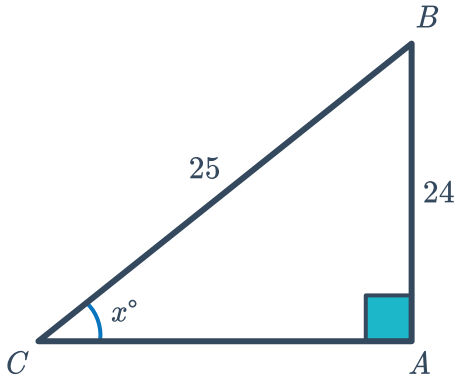
25 For each of the following, find θ to the nearest tenth of a degree:

a $\sin\theta = 0.8613$ b $\cos\theta = 0.4825$ c $\tan\theta = 8.958$ d $\sin\theta = 0.1234$

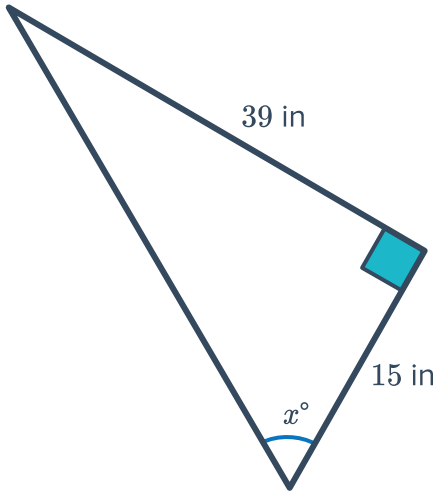
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26 Find the value of the variable, rounding your answer to the nearest degree:

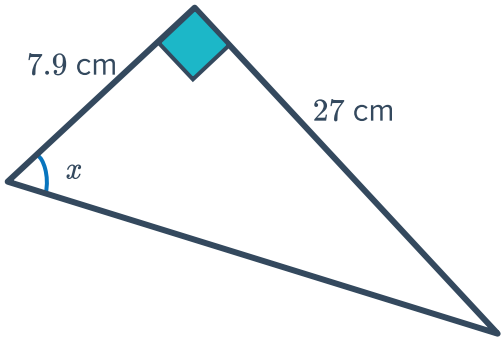
a



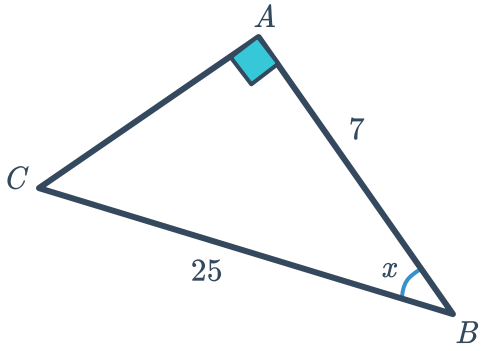
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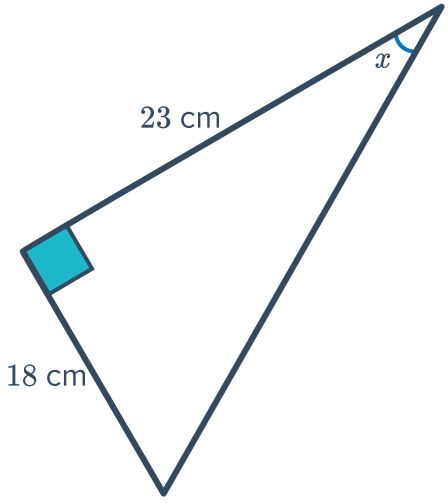
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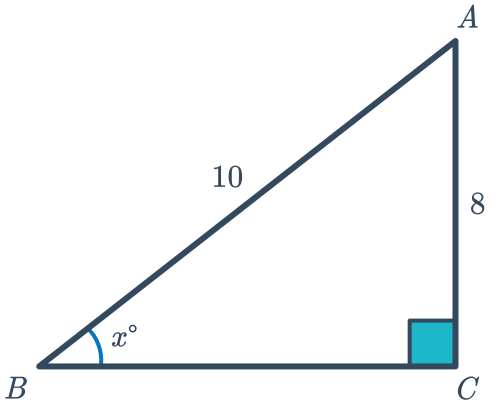
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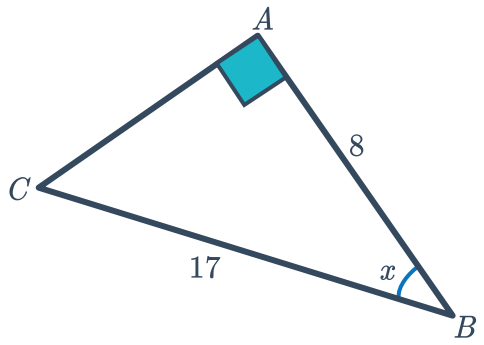


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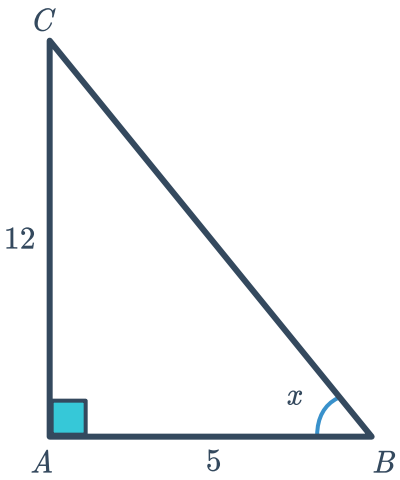


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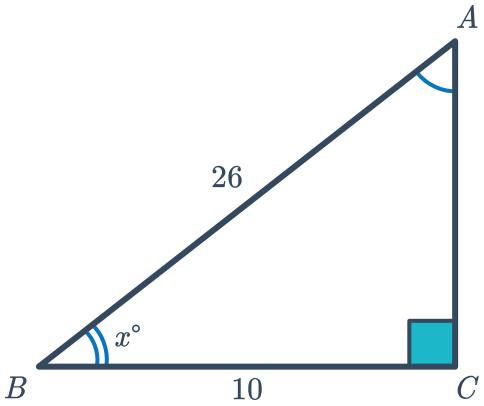
g



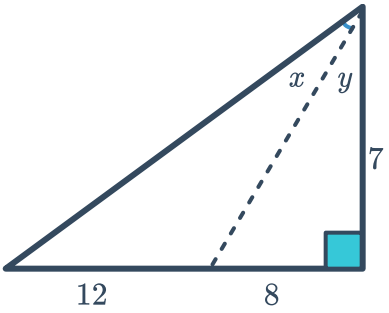
h



i

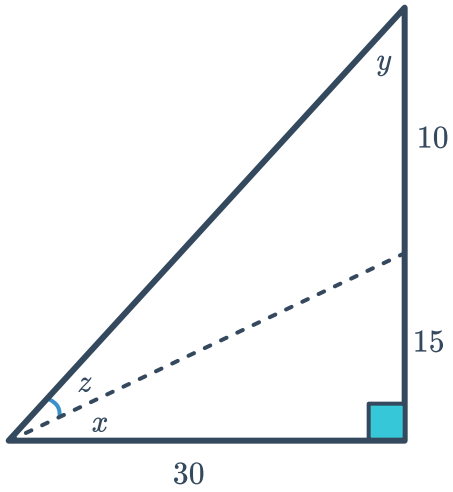


27 Consider the following diagram:
Find x and y, rounding your answers to one decimal place.



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28 Consider the given diagram:
Find the value of the variable, rounding your answer to two decimal places:

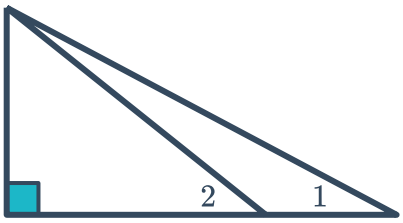


a x

b y

c z

29 Consider the following diagram:



a Which angle has the greater cosine value? Explain your reasoning

b Which angle has the greater sine value? Explain your reasoning

c Which angle has the greater tangent value? Explain your reasoning

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